

HiddenBench Imitation Control Study

Revised validation agenda, information measures, and weekly program

Working research note – 10 August 2026

1. Current hypothesis and what is now fixed

The current scientific object is a **reasoning-augmented multi-opinion imitation process with stochastic feedback control**. The design deliberately nests two established objects. HiddenBench supplies the semantic task: distributed evidence, a finite answer alphabet, and a known correct answer. The social-imitation model supplies the dynamical skeleton: one focal agent changes opinion at a time under an interaction-dependent stochastic rule.

For task r , let

$$\mathcal{A}_r = \{a_1, \dots, a_K\}, \quad Y^* \in \mathcal{A}_r$$

be the task options and ground-truth answer. Agent i has local evidence E_i and a committed opinion

$$X_i(t) \in \mathcal{A}_r.$$

The labelled microscopic state is

$$X_t = (X_1(t), \dots, X_N(t)),$$

while the mesoscopic occupation state is

$$\mathbf{n}_t = (n_1(t), \dots, n_K(t)), \quad p_a(t) = n_a(t)/N.$$

One elementary event changes at most one focal agent,

$$X_i(t) = a \rightarrow X_i(t+1) = b, \quad \mathbf{n}_{t+1} = \mathbf{n}_t - \mathbf{e}_a + \mathbf{e}_b.$$

The central experimental switch is:

reasoning ON: LLM update using HiddenBench evidence, memory, and local interaction,
 reasoning OFF: provider-free classical multi-opinion imitation update.

Everything else should be matched: task, option set, initial vote vector, population size, event geometry, controller target, controller sensing rule, horizon, and logging. The key scientific question is therefore not simply whether an LLM population can be controlled, but:

Does semantic reasoning change the macroscopic controllability of an imitation process?

2. State hierarchy and order parameters

Three state descriptions should be kept distinct:

$$X_t \longrightarrow \mathbf{n}_t \longrightarrow m_t.$$

X_t is agent-labelled and retains which evidence holder has which opinion. $\mathbf{n}_t$ is the primary *physics-level population state*. An order parameter m_t is a one-dimensional macroscopic projection of that state.

Let $Z \in \mathcal{A}_r$ be the controller target. Define

$$m_t^{\text{truth}} = \frac{Kp_{Y^*}(t) - 1}{K - 1}, \quad m_t^{\text{ctrl}} = \frac{Kp_Z(t) - 1}{K - 1},$$

$$m_t^{\text{ord}} = \frac{K \max_a p_a(t) - 1}{K - 1}, \quad \tilde{H}_t = -\frac{1}{\log K} \sum_a p_a(t) \log p_a(t).$$

Their interpretations are deliberately separate:

- m^{truth} : collective alignment with the correct answer;
- m^{ctrl} : collective alignment with the controller target;
- m^{ord} : collective ordering/consensus irrespective of which option wins;
- $\tilde{H}$: population-level disorder.

Evidence disclosure and evidence reach remain independent observables because HiddenBench’s characteristic failure is often failure to surface information, not inability to reason once the information is available.

A crucial point for the information analysis is that m_t^{ctrl} is an affine, one-to-one transform of the target count $n_Z(t)$ for fixed N and K . Therefore it is statistically safer to encode the discrete variable as $n_Z(t)$ while interpreting it as the order parameter m_t^{ctrl} . The same applies to m_t^{truth} and $n_{Y^*}(t)$.

3. The controller: stochastic measurement, feedback action, local actuation

The controller should be treated as a measurement-feedback loop,

$$\boxed{\mathbf{n}_t \longrightarrow Y_t \longrightarrow U_t \longrightarrow \mathbf{n}_{t+1}.$$

The population is the controlled system. The controller does not observe the full state perfectly. At event t , it samples q_c population agents without replacement and receives their opinion-count vector Y_t . Conditional on $\mathbf{n}_t$, this is a finite-population random measurement. Thus the controller has a genuine stochastic input even if its policy is deterministic.

The pilot policy can be

$$U_t = \pi_C(Y_t), \quad U_t \in \{\text{NO_OP}, \text{ADVOCATE_Z}\}.$$

For example, if sampled support for Z is below a threshold, the controller advocates Z ; otherwise it does nothing. Because Y_t varies, U_t can have nonzero entropy without introducing an arbitrary Bernoulli coin. A stochastic policy $\pi_C(u \mid Y_{\leq t})$ can be considered later if scientifically useful.

Actuation is local. In reasoning mode, `ADVOCATE_Z` replaces an ordinary local interaction by a fixed controller message; the focal LLM may accept or reject it. The controller must not overwrite the vote and does not receive hidden facts or the task rationale. In classical mode, the same control event modifies the transition weight toward Z in a provider-free way.

The controller must therefore satisfy three sanity properties before any information quantity is interpreted:

1. Y_t varies according to the intended finite-population sensor;
2. both control actions occur often enough that $H(U_t)$ is not approximately zero;
3. the controller changes transition statistics without directly forcing the focal state.

4. The information quantities we actually care about

The first analysis should deliberately distinguish **sensing**, **full population actuation**, and **order-parameter actuation**. This is the main revision to the earlier report.

4.1 System $\rightarrow$ controller: sensing information

The most complete low-dimensional sensing diagnostic is

$$I(\mathbf{n}_t; Y_t).$$

It asks how much the controller's measurement tells it about the current mesoscopic population state. Because Y_t is generated directly from $\mathbf{n}_t$, this quantity is also a calibration of the sensor implementation.

For control-relevant sensing, define the sampled target support $\hat{p}_Z(t)$, or its sampled order parameter $\hat{m}_t^{\text{ctrl}}$. Then

$$I(m_t^{\text{ctrl}}; \hat{m}_t^{\text{ctrl}})$$

asks how much the controller learns specifically about the macroscopic variable it is trying to regulate. Later, with history, this can be replaced by the appropriate transfer-entropy/directed-information form. The first pilot does not need to call this a stationary transfer-entropy *rate*.

4.2 Controller $\rightarrow$ population: mesoscopic actuation

The primary population-level actuation quantity is

$$\mathcal{T}_{U \rightarrow n} = I(U_t; \mathbf{n}_{t+1} \mid \mathbf{n}_t).$$

This asks: given where the population currently is, does knowing the control action improve prediction of where the entire occupation vector moves next? This is the closest first-pass quantity to the claim that the controller influences the *whole population dynamics*.

For $K > 2$, this contains more information than any single scalar order parameter. Two states can have identical target support but different distributions over the remaining options. Therefore $I(U_t; \mathbf{n}_{t+1} \mid \mathbf{n}_t)$ should be retained even when the headline interpretation uses an order parameter.

4.3 Controller $\rightarrow$ target order parameter: directed macroscopic control

The most interpretable quantity for the control objective is

$$\mathcal{T}_{U \rightarrow m^{\text{ctrl}}} = I(U_t; m_{t+1}^{\text{ctrl}} \mid m_t^{\text{ctrl}}).$$

Equivalently, for fixed N, K , estimate it from integer target counts,

$$I(U_t; n_Z(t+1) \mid n_Z(t)).$$

This asks exactly whether the controller action contains information about the next movement of the population *along the direction the controller is trying to steer*.

A complementary truth-directed quantity is

$$\mathcal{T}_{U \rightarrow m^{\text{truth}}} = I(U_t; m_{t+1}^{\text{truth}} \mid m_t^{\text{truth}}).$$

For a helpful controller with $Z = Y^*$, these two projections coincide. For a deliberately wrong target $Z \neq Y^*$, they separate control success from epistemic success. A strong adversarial controller may increase m^{ctrl} while decreasing m^{truth} .

4.4 Local diagnostic for sparse pilots

At very small sample sizes, conditioning on the full occupation vector can still be sparse. Keep the local fallback

$$I(U_t; X_{i_t}(t+1) \mid X_{i_t}(t), \mathbf{n}_t),$$

which asks whether control changes the focal transition after accounting for the focal opinion and population state. This should be treated as a diagnostic, not as the final population-level claim.

4.5 Mutual information is not control by itself

A positive conditional MI only says the controller action is predictive of subsequent dynamics. It does not say that the controller moves the system in the desired direction. Therefore every information quantity must be paired with a signed behavioral response, for example

$$R_{\text{ctrl}}(T) = \mathbb{E}[m_T^{\text{ctrl}} \mid \text{control}] - \mathbb{E}[m_T^{\text{ctrl}} \mid \text{no control}],$$

$$R_{\text{truth}}(T) = \mathbb{E}[m_T^{\text{truth}} \mid \text{control}] - \mathbb{E}[m_T^{\text{truth}} \mid \text{no control}].$$

A controller can have nonzero information transfer while merely perturbing the system. The signal of genuine control is therefore:

directed information + target-specific macroscopic displacement.

5. What the first experiments are meant to teach us

The initial experiment should be a 2×2 matched design:

	no control	feedback control
reasoning ON	A	B
reasoning OFF	C	D

All four cells use the same HiddenBench task, explicit initial vote vector, population size, event horizon, and controller target where applicable.

Each comparison answers a different question:

- $B - A$: can the controller steer an LLM reasoning population?
- $D - C$: can the same controller steer the classical imitation process?
- $A - C$: how does reasoning alter uncontrolled dynamics?
- $B - D$: how does reasoning alter *controlled* dynamics?

The first run is not intended to estimate a final phase diagram. It is intended to learn whether the experimental agenda itself is viable. Specifically, it should answer five questions:

1. **Does the sensor work?** Is $I(\mathbf{n}_t; Y_t)$ nonzero, above a shuffle null, and consistent with the expected finite-population sampling behavior?
2. **Does the controller actually actuate?** Does control change m^{ctrl} , target-adoption probability, or transition frequencies relative to the no-control cell?
3. **Is the actuation information-theoretically visible?** Are $I(U_t; \mathbf{n}_{t+1} \mid \mathbf{n}_t)$ and/or $I(U_t; m_{t+1}^{\text{ctrl}} \mid m_t^{\text{ctrl}})$ stable and above appropriate nulls?
4. **Is the effect target-specific?** If Z is reversed to a wrong answer, does the sign/direction of the macroscopic response follow Z rather than merely improving accuracy?
5. **Does reasoning matter?** Are response curves, transition kernels, evidence diffusion, or information-response relationships measurably different between reasoning ON and OFF?

6. Successful outcomes and failure diagnoses

The experiment has several levels of success. The first is engineering/statistical feasibility; the second is scientific promise.

6.1 Platform success

The platform is ready for serious experiments if:

- the classical mode makes zero provider calls and reproduces the intended one-agent jump process;
- the same explicit X_0 can be replayed in reasoning and classical modes;
- the stochastic sensor produces the expected variation in Y_t ;
- both `ADVOCATE_Z` and `NO_OP` occur with nontrivial frequency, so $H(U_t)$ is not degenerate;
- target-specific behavioral responses can be estimated from repeated episodes;
- the direct-counting MI/CMI estimators have transparent support diagnostics and sensible bootstrap uncertainty;
- shuffle/circular-shift nulls reduce the corresponding information signal.

6.2 Scientific green light

The strongest near-term success condition is:

- (i) feedback causes a reproducible signed shift in m^{ctrl} or target adoption,
- (ii) controller-to-population information survives the appropriate null,
- (iii) the magnitude or structure of this response differs between reasoning ON and OFF.

This would establish exactly the phenomenon the project needs: **reasoning changes macroscopic controllability**. It would justify scaling the system and investing in a richer history/semantic estimator.

A particularly strong result would be that reasoning changes the relationship between control information and response. For example, two systems may receive comparable control information but show different R_{ctrl} , or they may achieve comparable R_{ctrl} with different $\mathcal{T}_{U \rightarrow m^{\text{ctrl}}}$. That would motivate a later information-normalized control efficacy such as

$$\eta_{\text{info}} \sim \frac{R_{\text{ctrl}}}{\sum_t \mathcal{T}_{U \rightarrow m^{\text{ctrl}}}(t)},$$

while explicitly avoiding a thermodynamic-efficiency claim at this stage.

6.3 Diagnostic interpretation of negative outcomes

Observation	Interpretation / next move
$H(U) \approx 0$	Controller/sensor policy is degenerate. Change sensor size or threshold before doing MI work.
$X \rightarrow Y$ sensing works, but no behavioral response and no $U \rightarrow n$ signal	Measurement works; the actuation mechanism/controller message is ineffective. Redesign control, not the estimator.
Behavioral response exists, but CMI is unstable	Sampling/conditioning sparsity problem. Increase episodes or use target-count/focal projections before changing the game.
$U \rightarrow n$ CMI is above null, but m^{ctrl} does not move toward Z	Controller perturbs dynamics but does not achieve useful directed control.
Control response follows truth rather than controller target	Likely generic helpful-prompt/task-solving effect rather than target-specific control. Use wrong-target reversal.
Reasoning ON and OFF are indistinguishable	Semantic reasoning has not yet changed the relevant dynamics. Inspect evidence transmission/update behavior before scaling N .
Reasoning ON/OFF differ and target-directed information survives nulls	Strong green light. Freeze v1 and proceed to scale/structure experiments.

7. Which parameters to vary after the first signal

Order parameters are *measured outcomes*; they are not the main experimental axes. The first structural/control parameters are:

Axis	Symbol	Scientific role
Reasoning	ON/OFF	Primary comparison; vary immediately.
Population size	N	Finite-size/system-scale axis; canonical small- N first, then scale.
Sensor sample size	q_c	Measurement quality / controller observability.
Controller threshold	θ_c	Feedback aggressiveness / intervention frequency.
Controller target	Z	Correct vs wrong target; specificity and adversarial-control check.
Imitation interaction size	q	Later connection to nonlinear voter/q-voter structure.
Evidence allocation	scheme	Later test of redundancy, replication, and information fragmentation.

The first informative “control surface” after the pilot is likely (q_c, θ_c) , colored by R_{ctrl} , $\mathcal{T}_{U \rightarrow m^{\text{ctrl}}}$, or a response-per-bit quantity. Population size N should then become a finite-size extension. This ordering keeps the project focused: first establish that control exists and is measurable, then ask how control depends on sensing quality and system size.

8. Weekly program from the current point

Monday + Tuesday – finish and validate the experimental object. The game has now been implemented. Add the derived behavioral/controller metrics, verify all dynamical and privacy invariants, and run cheap classical/mock trajectories first. Then run the first real 2×2 grid with one validated task and a matched explicit X_0 . Do not optimize the controller and do not introduce InfoNCE.

Wednesday – test information-estimator feasibility. Adapt the existing direct-counting MI/CMI pipeline to the event table. Primary quantities are

$$I(\mathbf{n}_t; Y_t), \quad I(U_t; \mathbf{n}_{t+1} \mid \mathbf{n}_t), \quad I(U_t; n_Z(t+1) \mid n_Z(t)),$$

plus the focal fallback. Use episode-level bootstrap, permutation/circular-shift nulls, and contingency-support diagnostics. Wednesday’s output is an estimator-feasibility statement, not a paper claim.

Thursday – first scientific pilot. Run enough repeated real-provider episodes to compare the four cells. Plot per-option shares, m^{truth} , m^{ctrl} , m^{ord} , vote entropy, evidence diffusion, controller action markers, and the information estimates together. Add one wrong-target reversal if feasible.

Friday – go/no-go decision. Ask whether the combination of target-specific response, non-null directed information, and reasoning ON/OFF difference is present. If yes, freeze the game/controller v1. If behavior exists but MI is unstable, increase repetitions or simplify conditioning. If sensing works but actuation does not, redesign the controller. If ON/OFF are indistinguishable, inspect the semantic interaction rule before scaling.

9. Post-week extensions if the pilot is positive

The next extension should be **scale and structure**, not neural MI estimation immediately. Increase N using the already prepared replicated HiddenBench populations, while reporting time in normalized interaction units. Then vary one controller axis at a time, starting with q_c and θ_c . The central question becomes whether there are regimes in which control becomes easier, harder, or qualitatively different under semantic reasoning.

Only after the low-dimensional information signal is robust should the project move to history-aware semantic estimation using embeddings/InfoNCE. At that point, the methodological contribution is no longer “estimate MI from text” in the abstract; it is to estimate history-conditioned directed information in a nonstationary feedback language process.

The stochastic-thermodynamic extension comes later still. The classical Irisarri-like mode preserves the reaction-channel structure needed to explore microscopic reversibility and trajectory irreversibility. The LLM mode should not be assumed to satisfy those conditions. A controlled second-law-like statement is therefore a later theory objective, not a prerequisite for the current empirical agenda.

Bottom line. The immediate research question is now precise:

Does stochastic feedback produce a measurable, target-specific macroscopic control signal in Hidden-Bench imitation dynamics, and does semantic reasoning change the relationship between information acquired, information transmitted back to the population, and the resulting motion of the order parameters?

Anchoring sources. Y. Li, A. Naito, H. Shirado, “Systematic Failures in Collective Reasoning under Distributed Information in Multi-Agent LLMs,” ICML 2026. L. Irisarri, L. Trigo, R. Toral, G. Manzano, “Stochastic Thermodynamics of Social Imitation beyond Energetics,” 2026. J. M. Horowitz, H. Sandberg, “Second-law-like inequalities with information and their interpretations,” 2014. Current HiddenBench imitation implementation handoff and metrics/experiment implementation brief.